

1 Game Setting

Here I define a infinitely repeated game between consumer and producer.

- **Consumer**

Consumer has willingness to pay \$100.00 for a product, so his profit is the difference between his wtp and the price charged by the producer. Consumer's action space is $\{\pm 1\}$ where 1 represents take the producer's offer, while -1 represents reject the offer. Consumer's objective function is a sum of all expected future profits, i.e.

$$V_0(a) = \max_{a \in \{\pm 1\}} ((100 - x_0)\mathbb{I}(a = 1) + \beta E[V_1(a)])$$

the above is the consumer's bellman equation to be optimised where $\beta \in [0, 1]$ represents the time discount factor.

Consumer's policy :

$$\pi_t^c(a_t|x_t) = \epsilon \mathbb{I}(a_t = +1, x_t \leq 100) + (1 - \epsilon) \frac{1}{2} \mathbb{I}(a_t \in \{\pm 1\})$$

where $\begin{cases} \pi_*(a_t|x_t) = \begin{cases} 1 & x_t \leq 100 \\ -1 & \text{otherwise} \end{cases} \\ u_t(a_t, x_t) = (100 - x_t)\mathbb{I}(a_t = +1) \end{cases}$. Here the baseline (deterministic) policy is

$$\bar{\pi}(a_t|x_t) = \mathbb{I}(a_t = +1)\mathbb{I}(x_t \leq 100) + \mathbb{I}(a_t = -1)\mathbb{I}(x_t > 100)$$

- Information hiding (Reputation Theory)

: if consumer always accepts when $p \leq 100$, and refuses when $p > 100$, producer perfectly infers $w^* = 100$ and always charges $x = 100$ forever, giving consumer 0 surplus. Randomisation creates ambiguity.

- Exploration-Exploitation (Bandit Theory)

: consumer trades off immediate surplus (exploiting good prices) against long-term surplus (maintaining uncertainty). The optimal ϵ^* trades these off, similar to the UCB's exploration bonus.

Then we have a consumer's value function

$$V(x_t) = E_{x_{t+1} \sim \mu, a_t \sim \pi_t(\cdot|x_t)} [u_t(a_t, x_t) + \beta V_{t+1}(x_{t+1}) | \pi_t, x_t \sim \pi_t(\cdot|x_t)]$$

- **Producer**

Producer produces a product at the cost of \$50.00. The producer observes consumer's behaviour given each price and updates their belief about consumer's willingness to pay. Then they aim to extract full profit by setting the price as the consumer's willingness to pay.

Producer assumes consumer's willingness to pay is drawn from an unknown distribution. By the Central Limit Theorem, its mean follows approximately normal distribution: $\mathcal{N}(\mu_t, \sigma_t^2)$. The producer charges $x_t = \mu_t$ (the posterior mean) and updates (μ_t, σ_t^2) via Bayesian updates after each observation.

- at $t \in \{0, \dots, 6\}$

the producer explores consumer's choice by playing "take a middle" policy.

- $t \geq 7$

the producer uses data to fit the normal distribution and get the mean and set the price as the updated mean.

Here I denote by μ a measure for the producer's distribution for the states (the state here means the price the producer sets).

Producer's policy :

$$\pi_t^p(x_t) = \begin{cases} \pi' & 0 \leq t < \lfloor \log_2(H - L) \rfloor \\ \mu_t & t \geq \lfloor \log_2(H - L) \rfloor \end{cases}$$

where μ_t is the mean of normal distribution updated from the past t -observations, and π' is the optimal policy for a "guess a number" game;

Here I assume the range $[0, 150]$, hence $\lfloor \log_2(H - L) \rfloor = 7$. Thus, for time $t \in \{0, \dots, 6\}$, producer explores consumer's behaviour by playing "take the middle" policy. At $t = 7$, the producer fits data into the normal distribution with mean μ_7 and standard deviation σ_7 , i.e. $\mu \sim N(\mu_7, \sigma_7^2)$. After $t = 7$, the producer updates his/her distribution by Bayesian updates.

Algorithm 1 optimal policy π'

 $t = 0$ **while** True **do**For a interval $[L_t, U_t]$, $x_t = \lfloor (L_t + U_t)/2 \rfloor$ **if** $a_t = +1$ **then** $L_{t+1} = x_t$ **else** $U_{t+1} = x_t$ **end if****if** $L_{t+1} \neq U_{t+1}$ **then** $t+ = 1$ **else**

break

end if**end while**

2 Theoretical Foundations

2.1 Baseline Model

My baseline is that consumer plays truly to their preferences; if the price is higher than their willingness to pay, they reject it and conversely the price is lower than or equal to the willingness to pay, they accept it. Then it comes "guess-a-number" game to the producer, and they get maximum profits once they get the right number, while consumer only gets 0 surplus forever. In that game, the optimal policy is taking the middle number. In this setting, the producer sets price

$$(x_0, a_0) = (75, +1)$$

$$(x_1, a_1) = ((150 + 75)/2 \approx 112, -1)$$

$$(x_2, a_2) = ((112 + 75)/2 \approx 93, +1)$$

in range $[0, 150]$, the game will be $(x_3, a_3) = ((93 + 112)/2 \approx 102, -1)$ and af-

$$(x_4, a_4) = ((102 + 93)/2 \approx 97, +1)$$

$$(x_5, a_5) = ((102 + 97)/2 \approx 99, +1)$$

$$(x_6, a_6) = ((102 + 99)/2 = 100, +1)$$

ter trial 6, the producer's binary search converges to $w^* = 100$, and the producer will charge $p = 100$ forever from $t \geq 7$. In this case, the consumer's total profit is $25 + \beta^2 \cdot 7 + \beta^4 \cdot 3 + \beta^5 \cdot 1 + \beta^6 \cdot 0 + \sum_{i=7}^{\infty} \beta^i \cdot 0 = 25 + 7\beta^2 + 3\beta^4 + \beta^5$.

2.2 Noise Model

The base conjecture of this game is that randomisation obscures the signal so that the producer cannot observe the willingness to pay. This transforms the Baseline Model into a "guess with lies", and the producer side faces a noisy binary search problem.

Recall the deterministic baseline policy is $\bar{\pi}(a_t|x_t) = \mathbb{1}(a_t = +1)\mathbb{1}(x_t \leq 100) + \mathbb{1}(a_t = -1)\mathbb{1}(x_t > 100)$. A "lie" occurs when consumer deviates from it :

$\begin{cases} x_t \leq 100 \text{ but } a_t = -1 & \text{rejecting a good offer} \\ x_t > 100 \text{ but } a_t = +1 & \text{accepting a bad offer} \end{cases}$. Under such mixed policy, compute the conditional lie probabilities :
 $\begin{cases} \mathbb{P}(a_t = -1|x_t \leq 100) = \epsilon \cdot 0 + (1-\epsilon)(\frac{1}{2}) = \frac{1-\epsilon}{2} \\ \mathbb{P}(a_t = +1|x_t > 100) = \epsilon \cdot 0 + (1-\epsilon)(\frac{1}{2}) = \frac{1-\epsilon}{2} \end{cases}$. Denote $\alpha = \mathbb{P}(x_t \leq 100)$. Then the lie rate is

$$\begin{aligned} p &= \alpha \mathbb{P}(a_t = -1|x_t \leq 100) + (1-\alpha) \mathbb{P}(a_t = +1|x_t > 100) \\ &= \alpha \frac{1-\epsilon}{2} + (1-\alpha) \frac{1-\epsilon}{2} = \frac{1-\epsilon}{2} \end{aligned}$$

which implies the lie rate is independent of α , and therefore the noisy-search analysis becomes tractable.

Then based on noisy search theory [Karp2007], after t queries with lie rate p , posterior variance $\sigma_T^2 \geq C \exp(-TD(p))$ where $D(p) = (1/2 - p)^2$ is the discrimination rate. Then I have

$$\sigma_T^2 \geq C \exp(-T \cdot \epsilon^2/4) > 0$$

Therefore σ_T^2 is bounded below by a strictly positive constant depending on ϵ . This formally validates the strategy : randomisation prevents the producer's posterior from collapsing to a point mass at $w^* = 100$, ensuring positive expected consumer surplus over the infinite horizon. Explicitly, the consumer's value function is $V(x_t) = \mathbb{E}[(100-x_t)\mathbb{1}(a_t = +1) + \beta \cdot V_{t+1}]$ and $\mathbb{P}(a_t = +1) = \epsilon \mathbb{1}(x_t \leq 100) + (1-\epsilon)/2$. This optimisation problem can be solved iteratively by the tabular Q-learning algorithm.

3 Empirical Results

For further development, I have a bellman equation with continuous state (price) and discrete action (accept/reject). I took two approaches; Monte Carlo Simulation and Tabular Q-Learning.

3.1 Phase 1

In phase 1 (the first 0-6 trials), the producer plays the "take the middle" policy like they are in the "guess a number" game, while consumer randomises its behaviour with ϵ . Then we have a randomised version of state variable and consumer's behaviour as

$$\begin{aligned}
 (x_0, a_0) &= (75, \frac{1+\epsilon}{2}) \\
 (x_1, a_1) &= (\frac{149+75\epsilon}{2}, \frac{1-\epsilon^2}{2}) \\
 (x_2, a_2) &= (74.5 + 37.5\epsilon - 19\epsilon^2, \frac{2+\epsilon+\epsilon^3}{4}) \\
 (x_3, a_3) &= (\frac{594+336\epsilon-150\epsilon^2+36\epsilon^3}{8}, \frac{4+\epsilon-3\epsilon^2-\epsilon^3-\epsilon^4}{8}) \\
 (x_4, a_4) &= (\frac{-9\epsilon^4+26\epsilon^3-180\epsilon^2+346\epsilon+593}{8}, \frac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \\
 (x_5, a_5) &= (\frac{5\epsilon^5-11\epsilon^4+68\epsilon^3-364\epsilon^2+703\epsilon+1183}{16}, \frac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \\
 (x_6, a_6) &= (\frac{7\epsilon^5-7\epsilon^4+76\epsilon^3-368\epsilon^2+709\epsilon+1183}{16}, \frac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16})
 \end{aligned}$$

As a result, we reached somewhat a mixed-strategy equilibrium and the value for consumer at this phase becomes

$$\begin{aligned}
 V(\epsilon, \beta) &= \frac{-\beta^6\epsilon^5 + 45\beta^6\epsilon^4 + 38\beta^6\epsilon^3 - 462\beta^6\epsilon^2 - 37\beta^6\epsilon + 417\beta^6 + \beta^5\epsilon^5 + 49\beta^5\epsilon^4 + 46\beta^5\epsilon^3}{32} + \\
 &\frac{-250\beta^5\epsilon^2 - 31\beta^5\epsilon + 417\beta^5 + 4\beta^4\epsilon^5 + 54\beta^4\epsilon^4 + 64\beta^4\epsilon^3 - 420\beta^4\epsilon^2 - 20\beta^4\epsilon + 414\beta^4 + 12\beta^3\epsilon^4}{32} \\
 &+ \frac{8\beta^3\epsilon^3 - 424\beta^3\epsilon^2 - 8\beta^3\epsilon + 412\beta^3 + 56\beta^2\epsilon^3 - 296\beta^2\epsilon^2}{32} \\
 &+ \frac{56\beta^2\epsilon + 408\beta^2 - 408\beta\epsilon^2 + 408\beta + 400\epsilon + 400}{32}
 \end{aligned}$$

For the closed form solution as a function of β , I have the first order condition;

$$\begin{aligned}
 &-\frac{5}{32}\beta^6 + \frac{45}{8}\beta^6\epsilon^3 + \frac{57}{16}\beta^6\epsilon^2 - \frac{231}{8}\beta^6\epsilon - \frac{37}{32}\beta^6 + \frac{5}{32}\beta^5\epsilon^4 + \frac{49}{8}\beta^5\epsilon^3 + \frac{69}{16}\beta^5\epsilon^2 \\
 &\quad - \frac{225}{8}\beta^5\epsilon - \frac{31}{32}\beta^5 + \frac{5}{8}\beta^4\epsilon^4 + \frac{27}{4}\beta^4\epsilon^3 + 6\beta^4\epsilon^2 - \frac{105}{4}\beta^4\epsilon - \frac{5}{8}\beta^4 \\
 &\quad + \frac{3}{2}\beta^3\epsilon^2 - \frac{53}{2}\beta^3\epsilon - \frac{1}{4}\beta^3 + \frac{21}{4}\beta^2\epsilon^2 - \frac{37}{2}\beta^2\epsilon + \frac{7}{4}\beta^2 - \frac{51}{2}\beta\epsilon + \frac{25}{2} = 0
 \end{aligned}$$

but getting solution as a closed form requires unnecessary complexity. Instead, I plugged some different values of β in the above equation. As β differs from 0.5 to 0.99, ϵ^* varies from 0.57 to 0.07.

3.2 Phase 2

- Monte Carlo

I tabularised the continuous states (producer's distribution) and apply Monte-Carlo iteration to compute the value function and optimal ϵ for consumer's profits. For computation simplicity, I could not use the phase 1 data, instead assume $\mathcal{N}(100, 10^2)$. Then producer updates his belief about willingness to pay by Bayesian update with the observation of consumer's behaviour, and setting the new distribution's mean as a new price.

Algorithm 2 Monte Carlo Simulation

```

t,V=0,0
while  $t \leq 200$  do
  price=mean
   $r = \text{unif}(0,1)$ 
  if  $r < \epsilon$  then
     $a = +1$  if price $\leq$ wt $p$  else  $a = -1$ 
  else
     $a = \text{unif}(\{-1,+1\})$ 
  end if
  if  $a = +1$  then
     $V += \beta^t * (100 - \text{price})$ 
  end if
  update  $\mu, \sigma$  via Bayesian rule given ( $\text{price}, a$ )
   $t += 1$ 
end while

```

Here my pivotal argument is that if consumer is more forward-looking and therefore explore more, they earn more of surplus. It is derived by two results here. I first conjectured that myopic consumers (low β) prefer immediate optimal play over exploration, so ϵ should decrease with β . So I iterated with different time discount factor, and the result exactly shows the inverse correlation between them, the left of Figure 1. Also, when the consumer is more foresighted, in other words value more of future surplus, they get higher expected values, the right of Figure 1. These two empirical correlations confirm the theoretical hypothesis: patient consumers optimally trade immediate surplus for reputation, yielding higher long-run value.

- **Tabular Q-Learning**

As set in section 1, producer's belief is $\mathcal{N}(\mu_t, \sigma_t^2)$ and producer charges $x_t = \mu_t$. As the name suggests, we discretise it onto the grid $A_0 = \{0, 1, \dots, 150\}$ for tabular Q -learning so that the Q -table is indexed by observed states. Then we can compute posterior probability $\mathbb{P}(\text{wt}p = x' | \text{data}_t) = \mathcal{N}(x'; \mu_t, \sigma_t^2)$. The

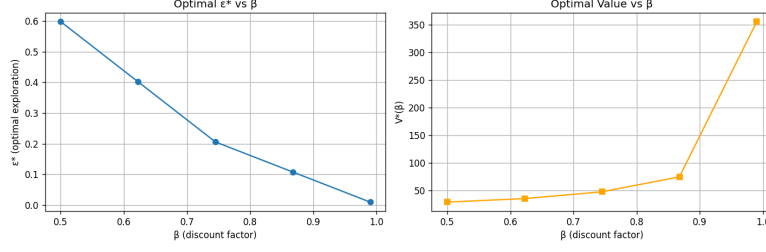


Figure 1: Correlations drawn from Monte Carlo

consumer applies Q-learning to learn optimal ϵ given observed prices. The Q-table implicitly captures expected future surplus through the Bellman backup. Firstly, define the Q-function for my problem

$$Q(x, \epsilon) = \mathbb{E}[u(x, \epsilon) + \sum_{j=1}^{\infty} \beta^j (100 - x_{t+j}) \mathbb{1}(a_{t+j} = +1) | x_t = x, \epsilon_t = \epsilon, \mu]$$

which has Bellman optimality equation

$$Q^*(x, \epsilon) = u(x, \epsilon) + \beta \sum_{x'} P_{xx'}(\epsilon) \max_{\epsilon'} Q^*(x', \epsilon'), \quad \forall x \in X, \epsilon \in \{0.0, 0.1, \dots, 1.0\}$$

With above equations, I can execute Q-learning update

$$Q_{t+1}(x_t, \epsilon_t) = (1 - \gamma) Q_t(x_t, \epsilon_t) + \gamma (u(x_t, \epsilon_t) + \beta \max_{\epsilon'} Q_t(x_{t+1}, \epsilon'))$$

where the learning rate is denoted as γ .

For state transition, I sample a fixed number of price states from the producer's posterior distribution, and build the Markov transition matrix based on that. As the game is repeated, this Markov transition matrix evolves.

CLAIM : There is a Bayesian Nash equilibrium which admits a stationary distribution.

Definition 3.1 (Carbonell-Nicolau, 2015). *A Bayesian game is a collection $\Gamma = ((T_i, \mathcal{T}_i), X_i, u_i, p)_{i=1}^N$ where $1, \dots, N$ is a finite set of players, $(T_i, \mathcal{T}_i)$ is a measurable space where T_i is player i 's type space, X_i is player i 's action space, $u_i : T \times X \rightarrow \mathbb{R}$ represents player i 's payoff function, and p is a probability measure on $(T, \otimes \mathcal{T}_i)$ denoting the common prior over type profiles.*

We define $i = 0$ (producer), $i = 1$ (consumer), with type spaces $T_0 = \{*\}$, $T_1 = \mathbb{R}_+$ (consumer's WTP), action spaces $A_0 = \{0, 1, \dots, 150\}$ (discretized prices),

$A_1 = \{0, 0.1, \dots, 1.0\}$ (discretised exploration rates), and stage payoffs:

$$\begin{aligned} u_0(t, x, \epsilon) &= \mathbb{E}_a[(x - 50)\mathbb{1}(a = +1)|x, \epsilon] = (x - 50) \cdot \mathbb{P}(a = +1|x, \epsilon) \\ u_1(t, x, \epsilon) &= \mathbb{E}_a[(t_1 - x)\mathbb{1}(a = +1)|x, \epsilon] = (t_1 - x) \cdot \mathbb{P}(a = +1|x, \epsilon) \end{aligned}$$

$a \sim \pi_\epsilon(\cdot|x)$ where $\mathbb{P}(a = +1|x, \epsilon) = (\epsilon)\mathbb{1}(100 \geq x) + (1 - \epsilon)/2$ The common prior p marginalizes to $t_1 \sim \mathcal{N}(100, \sigma_w^2)$.

Condition : $\forall i, \forall \delta > 0, \exists$ a $(\mathcal{B}(X_i), \mathcal{B}(X_i))$ -measurable map $\phi : X_i \rightarrow X_i$ s.t. the following holds ; $\forall (t, x) \in T \times X, \exists V_{x_{-i}}$ of x_{-i} s.t.

$$u_i(t, (\phi(x_i), y_{-i})) > u_i(t, (x_i, x_{-i})) - \delta, \forall y_{-i} \in V_{x_{-i}}$$

Proposition 3.1 (Carbonell-Nicolau, 2015). *If the Bayesian game satisfies the above condition, it is uniformly payoff secure.*

Verification of Condition (Proposition 3.1):

Both A_0 and A_1 are finite with discrete topology; for any action a , $\{a\}$ is open. Take $\phi_i = \text{id}$ for both players.

For producer ($i = 0$): $\forall (t, x, \epsilon)$, take $V_\epsilon = \{\epsilon\}$. Then

$$u_0(t, (\phi_0(x), \epsilon)) - u_0(t, (x, \epsilon)) = 0 > -\delta. \quad \checkmark$$

For consumer ($i = 1$): $\forall (t, x, \epsilon)$, take $V_x = \{x\}$. Then

$$u_1(t, (x, \phi_1(\epsilon))) - u_1(t, (x, \epsilon)) = 0 > -\delta. \quad \checkmark$$

By Proposition 3.1, Γ is uniformly payoff secure.

USC of sum: Since u_0 and u_1 are continuous on the discrete action space, both are USC. Their sum is USC.

Absolute continuity of prior: Since $T_0 = \{*\}$ is trivial, $p(t_0, t_1) = \delta_*(t_0) \cdot p_1(t_1)$, absolutely continuous w.r.t. the product of marginals.

Theorem 3.1 (Carbonell-Nicolau, 2015). *Suppose the Bayesian game $\Gamma = ((T_i, \mathcal{T}_i), X_i, u_i, p)_{i=1}^N$ is uniformly payoff secure, and $\forall t \in T, \sum_{i=1}^N u_i(t, \cdot) : X \rightarrow \mathbb{R}$ is upper semi continuous. If p is absolutely continuous w.r.t. $p_1 \otimes \dots \otimes p_N$, then there exists a Bayes-Nash equilibrium of Γ .*

Then by Theorem 3.1, this game has a Bayesian Nash Equilibrium.

We conjecture that stationary play of the stage BNE constitutes a Markov

perfect equilibrium of the discounted repeated game, with stationary joint distribution over $(\mu_t, \sigma_t^2, \epsilon_t)$. For the empirical validation, I simulate the game below. So even though the price state is not exactly Markov in the absence of producer's belief, we conjectured convergence under the BNE-induced stationary distribution.

Algorithm 3 Tabular Q-Learning

Input: β time discount factor, initial normal distribution p_0 , η convergence constant, T_{max} maximum iteration, learning rate γ , exploration rate ν .
Initialize Q_0 with zeros, $x_0 = \text{mean}(p_0)$, $t_0 = 0$.
while $t < T_{max}$ **do**
 Observe x_t and find $\epsilon_t^* = \arg\max Q_t(x, \epsilon)$
 With probability ν , play randomly from epsilon grid. With probability $1 - \nu$, play ϵ_t^* .
 Sample $a_t \sim \pi_{\epsilon^*}(\cdot | x_t)$, where $\pi_{\epsilon^*}(a = +1 | x_t) = \epsilon^* \mathbb{1}(x \leq 100) + (1 - \epsilon^*)/2$
 Observe $u_t(x_t, \epsilon_t^*) = (100 - x_t) \mathbb{1}(a_t = +1)$.
 update $\begin{cases} p_t \text{ with Bayesian update given } (x_t, a_t) \\ x_{t+1} = \text{mean}(p_{t+1}) \end{cases}$
 $Q_{t+1}(x_t, \epsilon_t^*) = Q_t(x_t, \epsilon_t^*) + \gamma(u_t(x_t, \epsilon_t^*) + \beta \max_{\epsilon'} Q_t(x_{t+1}, \epsilon') - Q_t(x_t, \epsilon_t^*))$
 if $\|Q_{t+1} - Q_t\|_\infty < \eta$ **then**
 Break
 end if
 $t = t + 1$
end while
return Q, π^*

With the exploration $\nu \neq 0$, and stationary state distribution (BNE conjecture), Q-learning algorithm empirically converges and therefore the above algorithm will give approximately true optimal policy. Also, here we only have the information about cumulative distribution function, not the point estimate, and therefore we use cdf to update the normal distribution (exact python code is provided at the github). The result gives no clear monotone correlation between time discount factor and optimal epsilon value. The optimal epsilon hovers around 0.4 to 0.6 across time discount factor β , suggesting state-adaptive optimal policy differs from fixed ϵ -optimum. Posterior σ remains bounded from 0 across all β , consistent with the noisy-search lower bound.

4 Conclusion

All empirical results coincide in the sense that they all show the benefits of randomisation: optimal $\epsilon^* > 0$, indicating randomisation dominates deterministic play in the consumer-producer game. The producer's posterior variance σ remains bounded above zero in all simulations, empirically validating the noisy-search lower bound.

The methods disagree on the precise function form of $\epsilon^*(\beta)$. By construction, Monte-Carlo grid-searches over fixed ϵ , while Q-learning learns state-adaptive ϵ^* . This discrepancy may cause this divergent trend.

Further work using different algorithm (e.g. Deep Q-Network) with richer state could consolidate the empirical characterisation of $\epsilon^*(\beta)$.

References

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